Pandas Data Cleaning and Visualization

Jupyter Notebook Export

Converted from the uploaded HTML notebook. The document includes the code cells, visible notebook outputs, and embedded charts available in the source file.

Notebook contents

Python data loading and inspection; duplicate and missing-value checks; text and numeric column cleaning; date conversion; descriptive statistics; and chart outputs embedded in the notebook.

Note: The uploaded notebook references a local CSV path on the original computer. That CSV file is not embedded in this HTML export.

Code Cell 1

```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
import seaborn as sns
df=pd.read_csv(r"C:\Users\Vykunta Reddy\Downloads\practise for pandas ecommerce1.csv")
print(df.head())
print(df.shape)
print(df.isnull().sum())
print(df.duplicated().sum())
df=df.drop_duplicates()
df.columns=df.columns.str.strip()
text_columns=[
    "Customer Name","City","Region","Segment","Product","Category","Payment Mode"
]
for col in text_columns:
    df[col]=df[col].astype("string").str.strip()
df["Region"]=df["Region"].str.title()
df["Product"]=df["Product"].str.title()
df["Payment Mode"]=df["Payment Mode"].str.title()
df["Order Date"]=pd.to_datetime(
    df["Order Date"],errors="coerce"
)
df["Sales"]=pd.to_numeric(
    df["Sales"].astype(str)
    .str.replace(r"[$,]","",regex=True),
    errors="coerce"
)
df["Quantity"]=pd.to_numeric(
    df["Quantity"].astype(str)
    .str.replace("units","",regex=False)
    .str.strip(),
    errors="coerce"
)
df["Unit Price"] = pd.to_numeric(
    df["Unit Price"], errors="coerce"
)
df["Profit"]=pd.to_numeric(
    df["Profit"],errors="coerce"
)
df["Discount"]=pd.to_numeric(
    df["Discount"],errors="coerce"
)
print(df.info())
print(df.isnull().sum())
print(df.duplicated().sum())
print(df[["Sales","Profit","Quantity"]].describe())
```

Output

```
Order ID  Order Date Customer ID Customer Name       City Region  \
0  ORD-1001  2024-02-02    CUST-108   Meera Patel  Bengaluru  North
1  ORD-1002  2024-09-25    CUST-175    Ananya Rao     Mumbai  South
2  ORD-1003  2024-10-12    CUST-141    Diya Reddy  Hyderabad   East
3  ORD-1004  2024-11-09    CUST-163   Rahul Kumar  Hyderabad   West
4  ORD-1005  2024-09-04    CUST-145  Aarav Sharma     Mumbai  North

       Segment       Product    Category Quantity  Unit Price  Discount  \
0    Corporate    Headphones  Technology        5        1800      0.10
1     Consumer    Headphones  Technology        6        1800      0.20
2  Home Office  Office Chair   Furniture        3        6500      0.20
3    Corporate    Headphones  Technology        2        1800      0.15
4  Home Office    Headphones  Technology        6        1800      0.05

     Sales   Profit Payment Mode
0   8100.0  1809.86  Net Banking
1   8640.0  -594.04  Net Banking
2  15600.0    894.9          UPI
3   3060.0    560.7  Net Banking
4  10260.0   684.22         Cash
(357, 15)
Order ID          0
Order Date        0
Customer ID       0
Customer Name     9
City             10
Region            0
Segment           0
Product           0
Category          0
Quantity          0
Unit Price        0
Discount          6
Sales             0
Profit            4
Payment Mode      0
dtype: int64
7
<class 'pandas.DataFrame'>
```

```
RangeIndex: 350 entries, 0 to 349
Data columns (total 15 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Order ID       350 non-null    str
 1   Order Date     345 non-null    datetime64[us]
 2   Customer ID    350 non-null    str
 3   Customer Name  342 non-null    string
 4   City           340 non-null    string
 5   Region         350 non-null    string
 6   Segment        350 non-null    string
 7   Product        350 non-null    string
 8   Category       350 non-null    string
 9   Quantity       350 non-null    int64
 10  Unit Price     350 non-null    int64
 11  Discount       344 non-null    float64
 12  Sales          343 non-null    float64
 13  Profit         345 non-null    float64
 14  Payment Mode   350 non-null    string
dtypes: datetime64[us](1), float64(3), int64(2), str(2), string(7)
memory usage: 65.9 KB
None
Order ID          0
Order Date        5
Customer ID       0
Customer Name     8
City             10
Region            0
Segment           0
Product           0
Category          0
Quantity          0
Unit Price        0
Discount          6
Sales             7
Profit            5
Payment Mode      0
dtype: int64
0
                Sales        Profit    Quantity
count     343.000000    345.000000  350.000000
mean    27022.959184   2163.104783    3.828571
std     54190.319966   8814.210614    1.962189
min       200.000000 -21558.910000   -3.000000
25%      1836.250000    -55.150000    2.000000
50%      5040.000000    230.080000    4.000000
75%     26650.000000   1012.900000    6.000000
max    315000.000000  87328.660000    7.000000
```

Code Cell 2

```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
import seaborn as sns
df=pd.read_csv(r"C:\Users\Vykunta Reddy\Downloads\practise for pandas ecommerce1.csv")
print(df.head())
print(df.shape)
print(df.isnull().sum())
print(df.duplicated().sum())
df=df.drop_duplicates()
df.columns=df.columns.str.strip()
text_columns=[
    "Customer Name","City","Region","Segment","Product","Category","Payment Mode"
]
for col in text_columns:
    df[col]=df[col].astype("string").str.strip()
df["Region"]=df["Region"].str.title()
df["Product"]=df["Product"].str.title()
df["Payment Mode"]=df["Payment Mode"].str.title()
df["Order Date"]=pd.to_datetime(
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df["Sales"]=pd.to_numeric(
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    .str.replace(r"[$,]","",regex=True),
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)
df["Quantity"]=pd.to_numeric(
    df["Quantity"].astype(str)
    .str.replace("units","",regex=False)
    .str.strip(),
    errors="coerce"
)
df["Unit Price"] = pd.to_numeric(
    df["Unit Price"], errors="coerce"
)
df["Profit"]=pd.to_numeric(
    df["Profit"],errors="coerce"
)
```

```
df["Discount"]=pd.to_numeric(
    df["Discount"],errors="coerce"
)
print(df.info())
print(df.isnull().sum())
print(df.duplicated().sum())
print(df[["Sales","Profit","Quantity"]].describe())
import matplotlib.pyplot as plt

# Chart 1: Average values
df[["Sales", "Profit", "Quantity"]].mean().plot(
    kind="bar", figsize=(8, 5)
)
plt.title("Average Sales, Profit and Quantity")
plt.ylabel("Average Value")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()

# Chart 2: Sales distribution
plt.figure(figsize=(8, 5))
plt.hist(df["Sales"].dropna(), bins=20, edgecolor="black")
plt.title("Distribution of Sales")
plt.xlabel("Sales Amount")
plt.ylabel("Number of Orders")
plt.tight_layout()
plt.show()

# Chart 3: Profit distribution
plt.figure(figsize=(8, 5))
plt.hist(df["Profit"].dropna(), bins=20, edgecolor="black")
plt.title("Distribution of Profit")
plt.xlabel("Profit")
plt.ylabel("Number of Orders")
plt.tight_layout()
plt.show()

# Chart 4: Quantity distribution
df["Quantity"].dropna().plot(
    kind="hist", bins=10, edgecolor="black", figsize=(8, 5)
)
plt.title("Distribution of Quantity")
plt.xlabel("Quantity")
plt.ylabel("Number of Orders")
plt.tight_layout()
plt.show()
```

Output

```
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Product           0
Category          0
Quantity          0
Unit Price        0
Discount          6
Sales             0
Profit            4
Payment Mode      0
dtype: int64
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 13  Profit         345 non-null    float64
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dtypes: datetime64[us](1), float64(3), int64(2), str(2), string(7)
memory usage: 65.9 KB
None
Order ID          0
Order Date        5
Customer ID       0
Customer Name     8
City             10
Region            0
Segment           0
Product           0
Category          0
Quantity          0
Unit Price        0
Discount          6
Sales             7
Profit            5
Payment Mode      0
dtype: int64
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min       200.000000 -21558.910000   -3.000000
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50%      5040.000000    230.080000    4.000000
75%     26650.000000   1012.900000    6.000000
max    315000.000000  87328.660000    7.000000
```

Chart 1

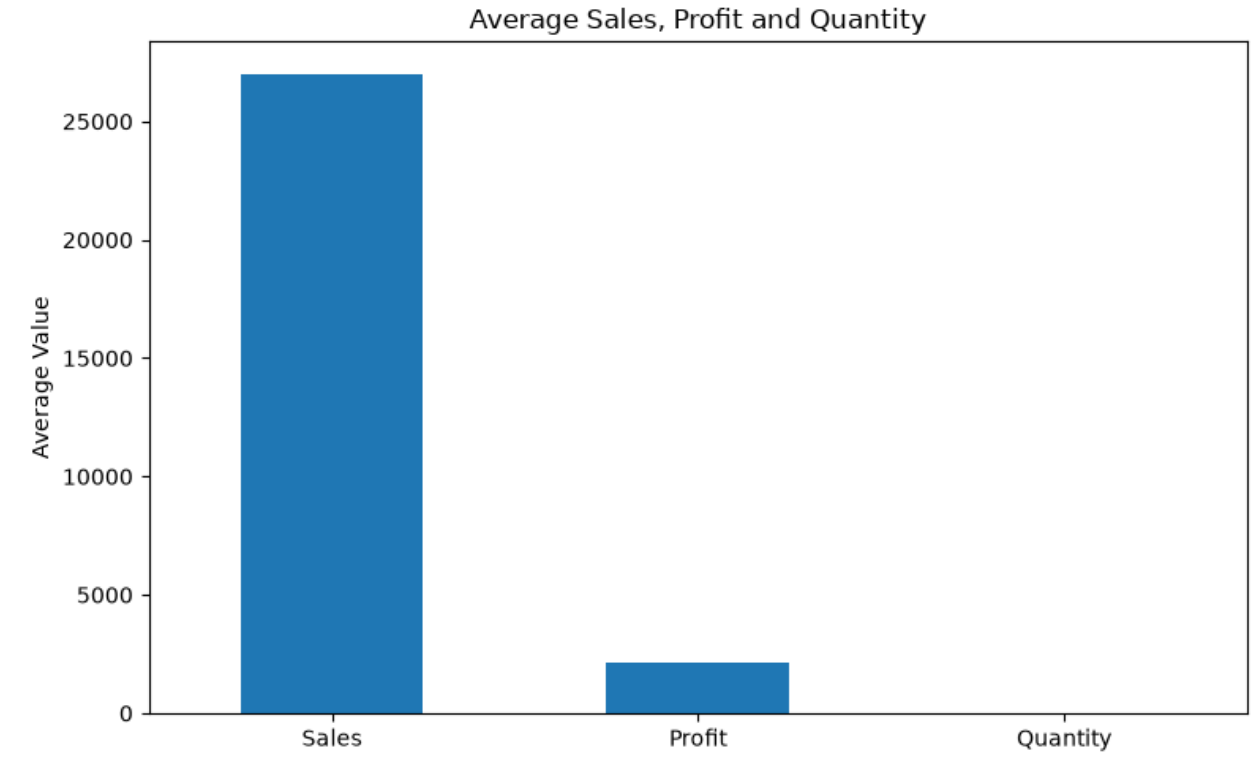


Chart 2

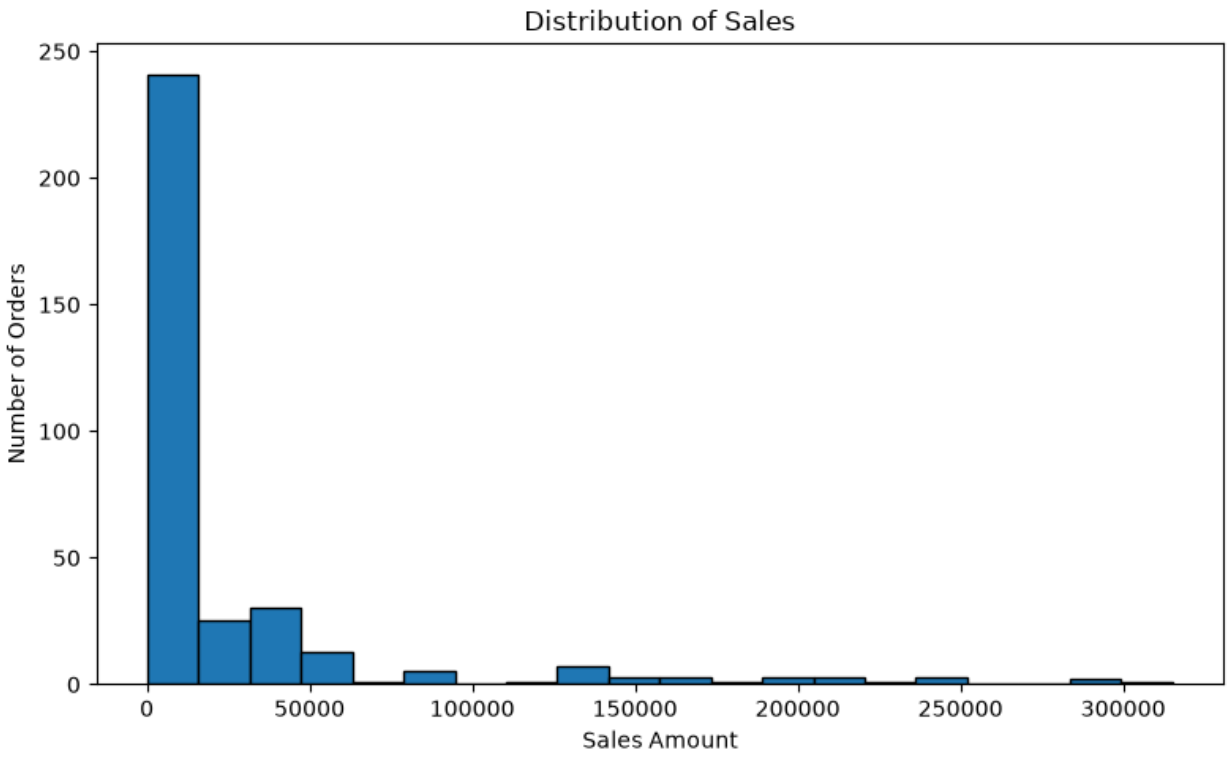


Chart 3

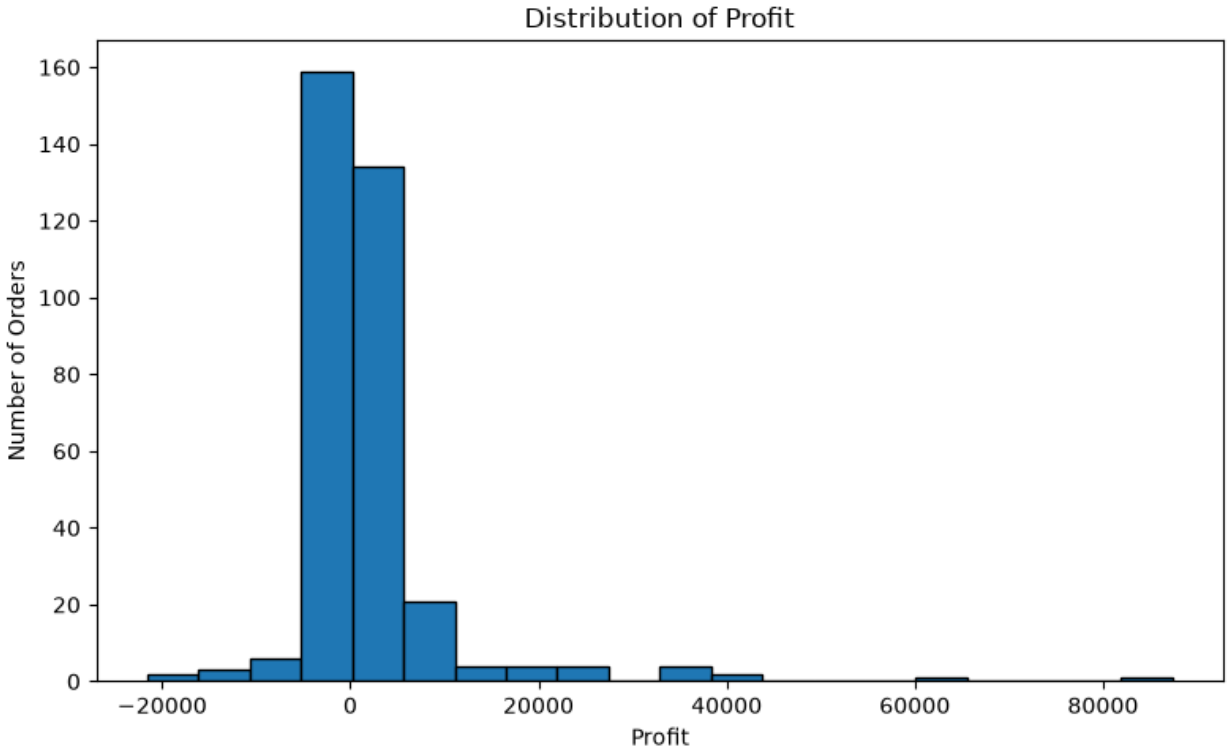
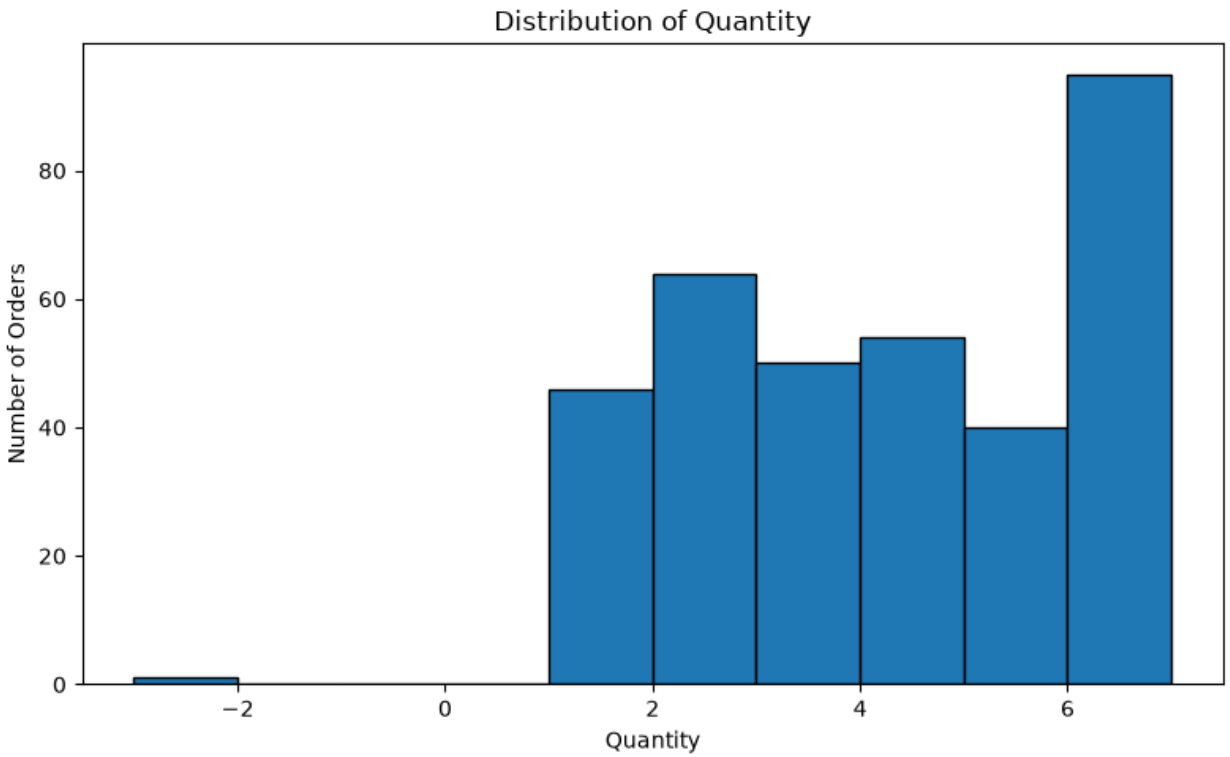


Chart 4


Distribution of Quantity
Number of Orders
0
20
40
60
80
−2
0
2
4
6
Quantity